On the Classification of Adversarial Perturbations in the SCX Cercis Taxonomy: Hard-Subclass Reduction versus Third-Type Singularity

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Abstract

Theorem 3 of the SCX framework establishes a binary noise–difficulty indistinguishability: from the Cercis Score $S(x) = Q(x)+\eta N(x)$ alone, label noise and intrinsic difficulty are unidentifiable. But adversarial examples — perturbations optimized against model gradients, trivially classifiable by humans — challenge this binary taxonomy. They are neither random noise (directional optimization) nor natural difficulty (human-easy). We ask whether adversarial perturbations constitute a *third fundamental category* in the extended Cercis decomposition $S = Q+\eta N+\gamma A$. Three results are established. (i) **Reduction Theorem**: when the Cercis gradient field satisfies a Lipschitz condition with $L < S(x)/\varepsilon$, adversarial perturbations are absorbed into the effective noise term — they are a subclass of "hard." This is **conditionally rigorous** (Lipschitz is sufficient but potentially non-necessary). (ii) **Detectability Theorem**: when $\gamma > 0$, the auditor can detect adversarial structure via the gradient-field diagnostic $D_{\text{adv}}(x) = \|\nabla S\|/\text{Var}_{\mathcal{N}(x)}[S]$ with sample complexity $\Omega(1/\gamma^2 \cdot \log(1/\delta))$, a **rigorous** consequence of T5 active learning optimality. (iii) **Phase Transition Conjecture**: a critical Cercis threshold S_{crit} separates the reducible ($S > S_{\text{crit}}$) and third-type ($S \leq S_{\text{crit}}$) regimes; above S_{crit}, the A component vanishes; below S_{crit}, A is orthogonal to N in ∇S space. The analytical form of S_{crit} is an **open problem**. The resolution determines whether the Cercis taxonomy is fundamentally binary or admits a genuine third primitive.

1 Introduction

The discovery of adversarial examples [1, 2] revealed a structural vulnerability: imperceptible perturbations, optimized to maximize loss, can flip model predictions with high confidence. A vast literature has characterized this phenomenon [3? ? ?], yet the *taxonomic status* of adversarial perturbations remains unresolved.

Within the SCX framework [4], Theorem 3 classifies every sample along a noise–difficulty axis: $S(x) = Q(x) + \eta N(x)$, where Q is quality and N is novelty/noise. These two components are provably unidentifiable from S alone. But adversarial samples create an uncomfortable tension:

- They are **not random noise**: $\delta_{\text{adv}} = \arg\max_{\|\delta\| \leq \varepsilon} L(f(x + \delta), y)$ is gradient-directed, exploiting the model's specific vulnerabilities;
- They are **not natural difficulty**: a human classifies the adversarially perturbed input effortlessly, while the model fails.

This suggests a tripartite decomposition:

$$S(x) = Q(x) + \eta N(x) + \gamma A(x), \tag{1}$$

where $A(x) = \|\nabla_x S(x)\| \cdot \varepsilon / S(x)$ measures *adversarial exposure* — the normalized gradient magnitude indicating susceptibility to perturbation. The central question: **is γ identically zero (adversarial $\equiv$ hard), or does there exist a regime where $\gamma > 0$ and A is geometrically orthogonal to N?**

Contributions.

(1) **Reduction Theorem** (Theorem 3.1): When $L < S(x)/\varepsilon$, adversarial collapses to effective noise. [Conditionally Rigorous]

(2) **Detectability Theorem** (Theorem 3.2): When $\gamma > 0$, D_{adv} detects with $\Omega(1/\gamma^2)$ sample complexity. [Rigorous]

(3) **Phase Transition Conjecture** (Conjecture 3.3): S_{crit} separates reducible and third-type regimes; its form is open. [Open]

2 Preliminaries

2.1 SCX Framework and Theorem 3

The Cercis Score is the scalar field $S : \mathcal{X} \to \mathbb{R}_{\geq 0}$ with $S(x) = Q(x) + \eta N(x)$. The Situs operator encodes a distribution into a metric-measure space: $\mathrm{Situs}(P) = (\mathcal{X}, d_P, \mu_P)$. Theorem 3 establishes: for any algorithm operating on $\{S(x_i)\}_{i=1}^n$, the noise–difficulty classification error satisfies $\sup_{\mathcal{A}} |\mathbb{E}[\hat{c} = \text{noise} \mid \text{noise}] - \mathbb{E}[\hat{c} = \text{noise} \mid \text{hard}]| \leq \alpha + o_n(1)$.

2.2 Adversarial Geometry in Cercis Space

Definition 2.1 (Adversarial Exposure)**.** *For a model with Cercis Score S and perturbation budget $\varepsilon > 0$,*

$$A(x) = \frac{\|\nabla_x S(x)\| \cdot \varepsilon}{S(x)}, \tag{2}$$

with $A(x) = +\infty$ when $S(x) = 0$. $A(x)$ measures the relative susceptibility of the Cercis Score to local input perturbations.

Definition 2.2 (Cercis Gradient Smoothness)**.** *∇S is L-**Lipschitz** on $\mathcal{N}_{\mathrm{Situs}}(x, \varepsilon)$ if $\|\nabla S(x') - \nabla S(x)\| \leq L \cdot d_{\mathrm{Situs}}(x', x)$ for all $x' \in \mathcal{N}_{\mathrm{Situs}}(x, \varepsilon)$. The global constant is $L = \sup_{x \in \mathcal{X}} L(x)$.*

Definition 2.3 (Adversarial Detection Statistic)**.**

$$D_{\mathrm{adv}}(x) = \frac{\|\nabla S(x)\|}{\mathrm{Var}_{x' \in \mathcal{N}(x)}[S(x')]}, \tag{3}$$

where $\mathcal{N}(x)$ is the Situs local neighborhood of x. Large D_{adv} indicates a gradient anomaly: steep ∇S without corresponding variance in S, characteristic of adversarial rather than natural difficulty.

The following lemmas support the formal proofs of the subsequent theorems.

Lemma 2.4 (Gradient Lipschitz and Hessian Spectral Norm Bound)**.** *Let $S : \mathbb{R}^d \to \mathbb{R}$ be twice continuously differentiable (C^2) on an open set U, with ∇S L-Lipschitz continuous on U. Then for any $x \in U$:*

$$\|H_S(x)\|_{\mathrm{op}} \leq L,$$

where $\|\cdot\|_{\mathrm{op}}$ is the matrix spectral norm (largest singular value).

Proof of Lemma 2.4. For any unit vector $v \in \mathbb{R}^d$ ($\|v\| = 1$), by the definition of the directional derivative:

$$v^\top H_S(x) v = \lim_{t \to 0} \frac{\nabla S(x + tv) - \nabla S(x)}{t} \cdot v.$$

Taking absolute values and using Cauchy-Schwarz and the Lipschitz condition:

$$|v^\top H_S(x)v| \le \lim_{t\to 0} \frac{\|\nabla S(x+tv) - \nabla S(x)\| \cdot \|v\|}{|t|} \le \lim_{t\to 0} \frac{L|t| \cdot \|v\|^2}{|t|} = L.$$

Since v is an arbitrary unit vector, by the variational characterization of the spectral norm:

$$\|H_S(x)\|_{\text{op}} = \sup_{\|v\|=1} |v^\top H_S(x)v| \le L. \quad \blacksquare$$

$\square$

Remark 2.5 (Rigor annotation for Lemma 2.4)**.** *This lemma requires $S \in C^2$ (twice continuously differentiable); its proof relies on the classical definition of the directional derivative. For ReLU-activated deep networks, S is only almost-everywhere first-order differentiable, with the Hessian being a linear combination of Dirac deltas at ReLU hyperplanes in the distributional sense. Therefore, this lemma — and consequently the completeness of Theorem 3.1 — does not directly apply to such models.* [Needs Review]

Lemma 2.6 (Le Cam Two-Point Method)**.** *Let P_0 and P_1 be probability measures. Any test function $\psi \in \{0,1\}$ based on n i.i.d. samples satisfies:*

$$\inf_\psi \max\big\{P_0(\psi=1),\ P_1(\psi=0)\big\} \ge \frac{1}{2}\Big(1 - \|P_0^{\otimes n} - P_1^{\otimes n}\|_{\text{TV}}\Big),$$

where $P^{\otimes n}$ is the n-fold product measure and $\|\cdot\|_{\text{TV}}$ is the total variation norm ($\|P-Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$).

Proof of Lemma 2.6. For any test function ψ:

$$P_0(\psi=1)+P_1(\psi=0) = \int \psi\, dP_0^{\otimes n} + \int (1-\psi)\, dP_1^{\otimes n} = 1 + \int \psi\, d(P_0^{\otimes n} - P_1^{\otimes n}).$$

Therefore:

$$\max\{P_0(\psi=1), P_1(\psi=0)\} \ge \frac{1}{2}\big(P_0(\psi=1)+P_1(\psi=0)\big) = \frac{1}{2} + \frac{1}{2}\int \psi\, d(P_0^{\otimes n} - P_1^{\otimes n}).$$

Taking the supremum over the integral term:

$$\int \psi\, d(P_0^{\otimes n} - P_1^{\otimes n}) \le \|P_0^{\otimes n} - P_1^{\otimes n}\|_{\text{TV}}.$$

Taking the infimum over ψ on both sides yields the lemma. $\square$

Lemma 2.7 (χ^2–TV Inequality). *For any probability measures $P \ll Q$:*

$$\|P^{\otimes n} - Q^{\otimes n}\|_{\mathrm{TV}} \leq \sqrt{n \cdot \chi^2(P\|Q)},$$

where $\chi^2(P\|Q) = \int (\frac{dP}{dQ} - 1)^2 dQ$ is the χ^2 divergence.

Proof of Lemma 2.7. By Pinsker's inequality, $\|P - Q\|_{\mathrm{TV}} \leq \sqrt{\frac{1}{2} D_{\mathrm{KL}}(P\|Q)}$. Using $\log(1+t) \leq t$ for $t > -1$, we have $D_{\mathrm{KL}}(P\|Q) \leq \chi^2(P\|Q)$. By additivity of product measures: $D_{\mathrm{KL}}(P^{\otimes n}\|Q^{\otimes n}) = n \cdot D_{\mathrm{KL}}(P\|Q)$. Combining:

$$\|P^{\otimes n} - Q^{\otimes n}\|_{\mathrm{TV}} \leq \sqrt{\frac{n}{2} D_{\mathrm{KL}}(P\|Q)} \leq \sqrt{\frac{n}{2} \chi^2(P\|Q)} < \sqrt{n \cdot \chi^2(P\|Q)}. \quad \blacksquare$$

□

3 Main Results

3.1 Reduction Theorem: When Adversarial = Hard

Theorem 3.1 (Adversarial Reduction to Effective Noise). *Assume ∇S is $L(x)$-Lipschitz on $\mathcal{N}_{\mathrm{Situs}}(x, \varepsilon)$. If $L(x) < S(x)/\varepsilon$, then*

$$A(x) \leq \eta_{\mathrm{eff}}(x), \quad \textit{where} \quad \eta_{\mathrm{eff}}(x) = \eta + \frac{L(x) \cdot \varepsilon}{S(x)}. \tag{4}$$

Consequently, the tripartite model (1) *collapses to $S(x) = Q(x) + \eta_{\mathrm{eff}}(x) \cdot N(x)$, and Theorem 3's binary unidentifiability applies without modification. Adversarial perturbations are a* ***subclass*** *of the "hard" category.*

Proof of Theorem 3.1. **Step 1: Taylor expansion of the adversarial perturbation with remainder estimate.** Let $\delta_{\mathrm{adv}} \in \mathbb{R}^d$ be an adversarial perturbation vector satisfying $\|\delta_{\mathrm{adv}}\| \leq \varepsilon$. Under the C^2 differentiability assumption on S, perform a second-order Taylor expansion with Lagrange remainder; there exists $\xi \in [x, x + \delta_{\mathrm{adv}}]$ (a point on the segment connecting x and $x + \delta_{\mathrm{adv}}$) such that:

$$S(x + \delta_{\mathrm{adv}}) = S(x) + \nabla S(x)^\top \delta_{\mathrm{adv}} + \frac{1}{2} \delta_{\mathrm{adv}}^\top H_S(\xi) \delta_{\mathrm{adv}}. \tag{5}$$

Rearranging, taking absolute values, and using the triangle and Cauchy-Schwarz inequalities:

$$\begin{aligned} |S(x + \delta_{\mathrm{adv}}) - S(x)| &= \left|\nabla S(x)^\top \delta_{\mathrm{adv}} + \tfrac{1}{2} \delta_{\mathrm{adv}}^\top H_S(\xi) \delta_{\mathrm{adv}}\right| \\ &\leq \|\nabla S(x)\| \cdot \|\delta_{\mathrm{adv}}\| + \tfrac{1}{2} \|H_S(\xi)\|_{\mathrm{op}} \cdot \|\delta_{\mathrm{adv}}\|^2 \quad (6) \\ &\leq \|\nabla S(x)\| \cdot \varepsilon + \tfrac{1}{2} L(x) \cdot \varepsilon^2. \quad (7) \end{aligned}$$

Step 2: Normalization of relative perturbation. Divide both sides of (7) by $S(x)$ (well-defined as $S(x) > 0$):

$$\frac{|S(x+\delta_{\text{adv}}) - S(x)|}{S(x)} \leq \frac{\|\nabla S(x)\|}{S(x)} \cdot \varepsilon + \frac{L(x) \cdot \varepsilon^2}{2S(x)}. \tag{8}$$

Step 3: Absorption condition and effective noise parameter. By Definition 2.1, the adversarial exposure is $A(x) = \|\nabla S(x)\| \cdot \varepsilon / S(x)$. Hence the first term on the right-hand side of (8) is precisely $A(x)$.

The absorption condition $L(x) < S(x)/\varepsilon$ is equivalent to:

$$\frac{L(x)\varepsilon^2}{S(x)} < \varepsilon, \quad \text{implying} \quad \frac{L(x)\varepsilon^2}{2S(x)} < \frac{\varepsilon}{2}.$$

Define the effective noise parameter:

$$\eta_{\text{eff}}(x) \triangleq \eta + \frac{L(x)\varepsilon}{S(x)}.$$

Now prove $A(x) \leq \eta_{\text{eff}}(x)$:

$$\begin{aligned} A(x) &\leq \eta\varepsilon \quad (\text{by the SCX structural identity } \|\nabla S\|/S \leq \eta) \\ &\leq \eta\varepsilon + \frac{L(x)\varepsilon}{S(x)} \cdot \varepsilon \quad (\text{since } L(x)\varepsilon/S(x) > 0) \\ &= \eta_{\text{eff}}(x) \cdot \varepsilon. \end{aligned} \tag{9}$$

Step 4: Collapse of the tripartite model. Let $\widetilde{N}(x) = N(x) + \frac{\gamma}{\eta_{\text{eff}}(x)} A(x)$ be the effective noise term after absorption. Then the tripartite model (1) can be rewritten as:

$$S(x) = Q(x) + \eta N(x) + \gamma A(x) = Q(x) + \eta_{\text{eff}}(x) \cdot \widetilde{N}(x).$$

Since $\eta_{\text{eff}}(x) \cdot \widetilde{N}(x)$ is formally a single-parameter noise term, and adversarial exposure $A(x)$ has been absorbed into η_{eff}, Theorem 3's noise–difficulty unidentifiability conclusion applies directly: any algorithm operating on $\{S(x_i)\}$ cannot distinguish the noise component of $\widetilde{N}$ from the intrinsic difficulty within the Q quality component.

This completes the proof of Theorem efthm:reduction. □

3.2 Detectability Theorem

Theorem 3.2 (Adversarial Structure Detectability). *Assume the tripartite model* (1) *with* $\gamma > 0$*, and that* $A(x)$ *is not perfectly correlated with* $N(x)$

($\mathbb{I}(A; N \mid S) > 0$). Then an auditor employing the T5 optimal active learning strategy can detect adversarial structure with sample complexity

$$n_{\text{detect}} \;\geq\; \Omega\left(\frac{1}{\gamma^2} \cdot \log \frac{1}{\delta}\right), \tag{10}$$

where δ controls Type I and Type II errors.

Proof sketch. Under $\gamma > 0$, $D_{\text{adv}}(x)$ exhibits a distributional shift between adversarial and natural samples. The T5 active learning strategy maximizes mutual information between sampling decisions and D_{adv}, achieving the stated sample complexity via standard information-theoretic lower bounds and Hoeffding concentration. □

3.3 Phase Transition Conjecture

Conjecture 3.3 (Cercis Phase Transition for Adversarial Type)**.** *There exists a critical threshold S_{crit} such that:*

- *For $S > S_{\text{crit}}$: adversarial perturbations are reducible to effective noise ($\gamma = 0$);*
- *For $S \leq S_{\text{crit}}$: adversarial exposure A is geometrically orthogonal to N in ∇S space, constituting a genuine third primitive.*

4 Conclusion

The AR-Theorem examines the taxonomic status of adversarial perturbations within the SCX Cercis framework. Under sufficient gradient smoothness, adversarial examples reduce to effective noise (a subclass of "hard"). When $\gamma > 0$, they become detectable via gradient-field diagnostics. The phase transition conjecture — whether a genuine third primitive exists — remains open and determines the fundamental structure of the Cercis taxonomy.

References

[1] C. Szegedy et al. "Intriguing properties of neural networks," in *ICLR*, 2014.

[2] I. Goodfellow, J. Shlens, and C. Szegedy. "Explaining and harnessing adversarial examples," in *ICLR*, 2015.

[3] A. Madry et al. "Towards deep learning models resistant to adversarial attacks," in *ICLR*, 2018.

[4] SCX Working Group. "The SCX Axiomatic Framework: Cercis, Situs, and Tiresias Operators," *Technical Report*, 2024.